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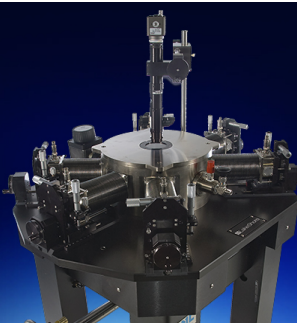
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ABSTRACT

In this paper, we present the proof-of-concept of a device for rectifying ac-magnetic-field effects by mixing thermal spin and charge currents. The device consists of a NiO/Pt bilayer in which a pure spin current is generated perpendicularly to the sample plane by means of the spin Seebeck effect. The pure spin current, created in the NiO layer, is injected into the Pt layer and is converted into a charge current by means of the inverse spin Hall effect, which is measured by electrical contacts. Due to the combination of two effects, the spin Seebeck and the planar Nernst effect, the overall response of the device as a function of a magnetic field is similar to the one of a rectifying diode. The resulting charge current mainly flows in one direction, approaching zero for negative values of the magnetic field when the spin Seebeck voltage has an opposite sign and magnitude close to the planar Nernst effect contribution. This remarkable effect is well explained by analyzing the response function as well as taking into account the magnon diffusion theory for the spin Seebeck effect. We believe that this proof of concept is of interest to the development of spintronic devices.

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The discovery of pure spin current generation and manipulation in magnetic insulators is one of the most remarkable advances in spintronics. As the movement of electron charge results in unavoidable joule heating in metals, the pure spin current with no charge transportation has received great attention lately. Therefore, the search for materials and methods of pure spin current generation and manipulation in insulators is of current interest. Among the main ways of spin current generation, we mention the spin Hall effect (SHE)¹ and the spin Seebeck effect (SSE).² In the SHE, a charge current flowing in nonmagnetic metals with a large spin-orbit interaction (SOI) generates a transverse spin current. In the SSE, a temperature gradient across a magnetic material generates a spin current, whose propagation direction depends on the experimental geometry. It can be perpendicular or parallel to the temperature gradient, characterizing the transversal (TSSE) or longitudinal (LSSE) spin Seebeck effects, respectively. While the TSSE can be observed in both magnetic metals and magnetic insulators, the LSSE is only observed in magnetic insulators, because it is free of the anomalous Nernst effect.^{3,4} In magnetic insulators, the thermal gradient creates a longitudinal nonuniform spin accumulation that results in a spin-current carried by spin waves or magnons.⁵ By depositing a heavy metal layer on the magnetic material,

the spin current is injected through the interface and is converted into a transverse charge current by means of the inverse spin Hall effect (ISHE).^{6–8}

In order to advance spintronics toward integration with conventional electronic circuits, much effort has been put in the development of circuitry elements such as spin-batteries,^{9,10} spin-transistors,^{11,12} and spin-diodes.^{13,14} Most of these device proposals are based on ferromagnetic materials that are difficult to function in nanoscale sizes due to the stray field distribution that establishes electromagnetic links with nearby devices. Therefore, the search for devices based on antiferromagnetic (AF) materials, which do not have the stray field trouble, is a challenging task for basic and applied spintronics investigation. Antiferromagnetic materials are being considered as storage media either using the spin-orbit torque effect¹⁵ or using anisotropic magnetoresistance in the AF metal FeRh.¹⁶ Certainly, AF materials present intrinsic magnetic properties that make them very exciting for spintronics applications.^{17–25} For instance, nickel oxide (NiO) is one of the best options among the AF materials for development of spintronic devices. Thin films of NiO are easily prepared by conventional techniques such as sputtering and Pulsed Laser Deposition (PLD); it has a Néel temperature (T_N) well above room temperature ($T_N = 523$ K), it

presents a relative small in-plane magnetic anisotropy, and it has a spin-flop magnetic field at a room temperature of about 52 kOe.²⁶

In this paper, we show that by combining thermal spin-current and charge-current effects in a NiO/Pt nanostructure, one can build a device for the rectifying spin-Seebeck-effect (SSE) modulated by ac-magnetic fields or ac-spin currents injected across the NiO/Pt interface. As we show below, the device works as a diode type rectifying transducer for ac spin-currents generated by modulating the thermal gradients.^{27,28} While previous investigations of spin diode systems are mainly focused on metallic spin-valves, which are based on the spin transfer torque (STT), the results reported here are among the first ones to show the rectifying effect for spin-currents with no charge transportation.

A sketch of the spin current-to-charge current conversion diode, which is based on the SSE driven by an ac-magnetic field, is shown in Fig. 1(a). The sample basically consists of a bilayer of NiO (100 nm/Pt(6 nm) sputtered on commercial Si(100) with a buffer layer of Ta(3 nm) and a thin underlayer of permalloy (Py). As explained in Ref. 29, the role played by the Py layer is only to induce antiferromagnetic texture in NiO. The temperature gradient is applied along the y-direction, normal to the xz -plane of the sample (1.5 mm $\times$ 0.5 mm), with an in-plane dc magnetic field applied along the z direction, in the LSSE configuration. The Pt layer, used to convert the pure spin current in charge current, is a four-terminal Hall-bar deposited on top of the NiO layer by means of a shadow mask, which is 1.5 mm long and 0.5 mm wide. A 51 Ω SMD (surface mount device) resistor was used as a heater, while CERNOX thermometers (model CX-1050-SD-HT) were used to determine the perpendicular thermal gradient temperature $\vec{\nabla}T_y$. It is interesting to notice that a single heater attached to a copper film produces the transverse in-plane ($\vec{\nabla}T_z$) and the perpendicular ($\vec{\nabla}T_y$) thermal gradients as sketched in Fig. 1(b).

Due to the thermal gradient normal to the plane, a longitudinal spin-current density $\vec{J}_S$ is induced in the NiO film, which is given by $\vec{J}_S = -C_s \vec{\nabla}T_y$, where C_s is a parameter that depends on the thermal populations and on the dispersion relations of the acoustic magnons associated with the two magnetic sublattices of the AF-NiO, on the

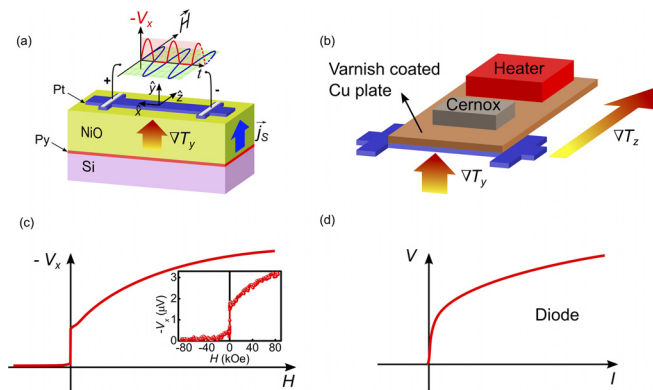


FIG. 1. (a) Spin-current rectification device. The in-plane ac magnetic fields combined with the perpendicular and in-plane temperature gradients shown in (b) act in order to create a singular overall response $-V_x \times H$ as sketched in (c). The actual experimental data are shown in the inset. The H -dependence of negative values of V_x has some similarities with the $V \times I$ curve for a diode as shown in (d).

applied magnetic field (H), and on the temperature (T).³⁰ $\vec{J}_S$ is actually detected after being converted into a transversal charge current $\vec{J}_C$ in the Pt-film by means of the inverse-spin-Hall effect, where $\vec{J}_C = \theta_{SH} \vec{J}_S \times \hat{\sigma}$. In this expression, θ_{SH} is the spin-Hall angle that measures the efficiency in the spin-to-charge current conversion and $\hat{\sigma}$ is the polarization of the spins determined by the direction of H . Finally, the longitudinal voltage (V_x) measured along the Hall bar of Pt is a combination of two effects: $\vec{J}_C$ due to the spin-Seebeck voltage (V_{SSE}) and the planar Nernst effect (V_{PNE}). V_x was actually detected by the charge accumulation in the Pt Hall bar electrical contacts by using a dc-resistivity puck of a PPMS (Physical Properties Measurement System made by Quantum Design) in an open circuit configuration.³⁰ As shown in Fig. 1(a), the magnetic field value applied along the z -axis was sinusoidally varied as a function of time as $H(t) = H_0 \sin \omega t$, where $H_0 = 40$ kOe, and $\omega \cong 0.166$ rad/min, which corresponds to a period of 40 min. The somewhat low frequency of the applied ac-magnetic-field was mainly limited by the large inductance of the 9 T superconducting solenoid which allows one to vary the magnetic fields at rates smaller than 100 Oe/s. However, the magnetization of the AF-NiO is known to follow the magnetic field in much higher rates and the proposed device is not limited to low frequencies only. By keeping the temperature gradient fixed and varying the magnetic field value, the longitudinal voltage is measured along the Pt Hall bar and an asymmetrical response is obtained as illustrated in Fig. 1(c). For negative values of H , $-V_x$ is constant, and as H crosses the zero field value, V_x suddenly jumps for a substantial value for $H \geq 0$ Oe. The response function of the $-V_x \times H$ resembles the $V \times I$ curve of a diode [shown in Fig. 1(d)]. Therefore, the device works as a half-wave rectifier.

Figure 2(a) shows the V_x data measured at room temperature for H varying in the range ± 85 kOe without (red symbols) and with (blue symbols) a temperature gradient. With no temperature gradient, the $-V_x \times H$ response is constant (red symbols), indicating that no spin current is being injected across the NiO/Pt interface. However, for $\Delta T = 10$ K, we measured a highly asymmetric $V_x \times H$ dependence, as shown by the data in blue and shown in the inset of Fig. 1(c) where the offset was subtracted. In Fig. 2(a), the V_x signal approaches to the one for $\Delta T = 0$ when $H < 0$ and departs significantly for positive values of H . Due to the $-V_x \times H$ response, an ac-magnetic field applied along the z -direction modulates the spin Seebeck effect so that it generates a charge current along the x -direction only for positive values of H , working as a spin-current rectifier. In order to understand the

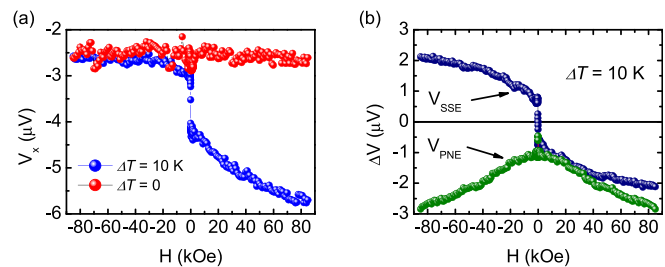


FIG. 2. (a) The H -dependence for the overall $V_{SSE} + V_{PNE}$ signal measured at room temperature by applying a thermal gradient of 10 K (blue) and in the absence of a thermal gradient (red). The V_x signal is a combination of an asymmetric contribution due to the spin-Seebeck effect (dark blue) and of a symmetric planar Nernst effect (green) as shown in (b).

rectification effect, the V_x data can be represented as a sum of two components, $V_x = V_{SSE}^S(H) + V_{SSE}^A(H)$, where the symmetric component is characterized by $V_{SSE}^S(-H) = V_{SSE}^S(H)$ and the asymmetric component by $V_{SSE}^A(-H) = -V_{SSE}^A(H)$. As H reverses its signal, the spin current also reverses its orientation. Therefore, the asymmetric component is generated by the spin Seebeck voltage. On the other hand, the origin of the symmetric component is more involved. As the Pt atoms at the NiO/Pt interface get a magnetic polarization due to the proximity effect, there occurs a planar Nernst effect that is symmetrical as a function of the magnetic field.^{31,32} After separating the symmetrical and asymmetrical components, they can be plotted as a function of the magnetic field as shown in Fig. 2(b). Notice that the asymmetric component (blue symbols) resembles a magnetization curve, while the symmetric component (green symbols) is characteristic of the planar Nernst effect as discussed in Ref. 32. As shown in Fig. 2(b), the difference in the symmetries of both thermal effects allows one to separate them from each other.

By removing the offset of the H -dependent signal shown in Fig. 2(a) (blue data), we obtain a $-V_x \times H$ curve similar to the I - V curves of rectifying diodes, as shown in Fig. 1(d). The offset voltage is actually due to thermo-electrical voltages commonly observed in two lead dc transport measurements and to instruments used in these kinds of measurements. Therefore, the device shown in Fig. 1(a) can be used for rectifying ac-signals originated from the pure-spin conversion in charge-current. The rectification property of our spintronic device is shown in Fig. 3, where the amplitude of the ac-magnetic field is 40 kOe (blue data) and the V_x response shows a typical half-wave rectifying signal (red data). The response of the rectifying device was also measured by varying the amplitude of the ac-magnetic field as well as the base temperature. The results shown in Figs. 4(a) and 4(b) were obtained with $\Delta T = 10$ K, and the values are relative to the smaller value of the signal detected by the device. The dependence of V_x with H follows the one expected from the data shown in Fig. 2, null for $H < 0$ and increasing tendency for $H > 0$, as shown in the inset of Fig. 1(c). However, the dependence of V_x as a function of T for $\Delta T = 10$ K and keeping the amplitude of the ac magnetic field in 85 kOe varies more dramatically as shown in Fig. 4(b). As discussed in Refs. 29 and 30, the spin Seebeck coefficient (S_{SSE}) is defined as

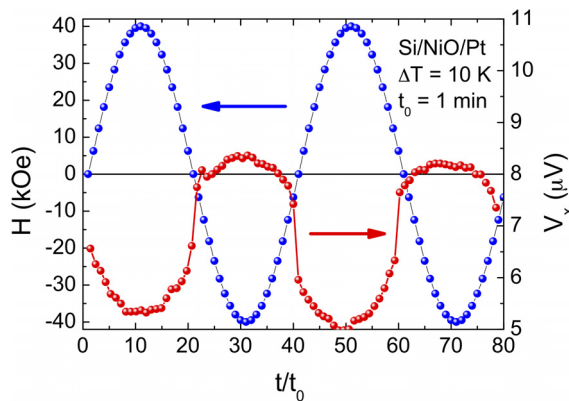


FIG. 3. Time-response of the spin-Seebeck rectified signal (red) measured for a base temperature of 290 K and modulating magnetic field with an amplitude of 40 kOe (blue) for $\Delta T = 10$ K.

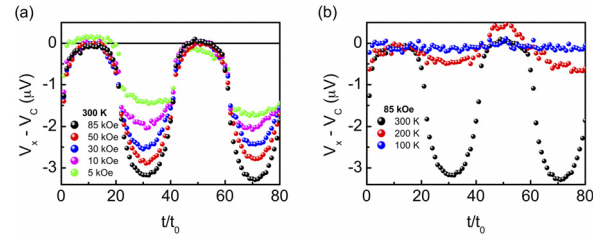


FIG. 4. (a) Time-response of the rectified signal modulated by ac magnetic fields with amplitudes of 5, 10, 30, 50, and 85 kOe and for $T = 290$ K and (b) the time response measured for $T = 100, 200$, and 300 K with an amplitude of the ac magnetic field of 85 kOe.

$S_{SSE} = V_{SSE} / \left(R_{Pt} \left| \vec{\nabla} T_y \right| \right)$, where R_{Pt} is the Pt resistance and $\vec{\nabla} T_y$ is the temperature gradient perpendicular to the NiO layer. According to the magnon diffusion model developed in Ref. 30, the spin Seebeck coefficient is given by

$$S_{SSE} = w \lambda_N \frac{2e}{\hbar} \theta_{SH} \tanh\left(\frac{t_N}{2\lambda_N}\right) C_S, \quad (1)$$

where w , t_N , λ_N , and θ_{SH} are the width, thickness, spin diffusion length, and spin Hall angle of the Pt layer, respectively. The coefficient C_S is explicitly calculated for the NiO/Pt bilayer and is given by Eq. (64) of Ref. 30. The temperature dependence of S_{SSE} plotted in Fig. 4(b) of Ref. 30 shows its dramatic decrease in the low temperature regime, which is due to the decrease in the magnon thermal population, in agreement with our result depicted in Fig. 4(b).

In summary, we reported the very exciting phenomenon of the ac spin current-to-charge current rectification in a NiO/Pt bilayer. In the present work, the ac-magnetic-field was actually used for modulating the spin-to-charge conversion by switching the direction of the magnetization and not the direction of the spin-current. However, due to the symmetry of the mechanisms involved, by fixing H and thermal modulating the spin-current, the spin-to-charge conversion will also be modulated in a similar manner as observed when using ac-magnetic-fields. The rectifying behavior is due to two combined effects: the magnetic field dependence of the spin Seebeck effect plus the planar Nernst effect (PNE). The symmetric behavior of the SSE and the asymmetric behavior of the PNE, acting simultaneously create an overall response of our device that is similar to a conventional rectifier diode. The field dependence of the rectified V_x signal is fully understood in terms of the measured response function of the device. On the other hand, the decrease in the V_x signal in the low temperature regime is also well explained by the magnon diffusion theory. We are confident that this is a very stimulating proof of concept that explores the idea of how to implement a spin current-to-charge current diode and it must be of interest for spintronics applications.

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REFERENCES

- ¹J. Sinova, S. O. Valenzuela, J. Wunderlich, C. H. Back, and T. Jungwirth, *Rev. Mod. Phys.* **87**, 1213 (2015).
- ²K. Uchida, S. Takahashi, K. Harii, J. Ieda, W. Koshibae, K. Ando, S. Maekawa, and E. Saitoh, *Nature* **455**, 778–781 (2008).
- ³K. Uchida, T. Nonaka, T. Ota, H. Nakayama, and E. Saitoh, *Appl. Phys. Lett.* **97**, 262504 (2010).
- ⁴B. F. Miao, S. Y. Huang, D. Qu, and C. L. Chien, *Phys. Rev. Lett.* **111**, 066602 (2013).
- ⁵S. M. Rezende, R. L. Rodríguez-Suárez, R. O. Cunha, J. C. López Ortiz, and A. Azevedo, *J. Magn. Magn. Mater.* **400**, 171–177 (2016).
- ⁶A. Azevedo, L. H. Vilela-Leão, R. L. Rodríguez-Suárez, A. B. Oliveira, and S. M. Rezende, *J. Appl. Phys.* **97**, 10C715 (2005).
- ⁷E. Saitoh, M. Ueda, H. Miyajima, and G. Tatara, *Appl. Phys. Lett.* **88**, 182509 (2006).
- ⁸A. Hoffmann, *IEEE Trans. Magn.* **49**, 5172 (2013).
- ⁹A. Brataas, Y. Tserkovnyak, G. E. W. Bauer, and B. I. Halperin, *Phys. Rev. B* **66**, 060404(R) (2002).
- ¹⁰J. Tian, S. Hong, I. Miotkowski, S. Datta, and Y. P. Chen, *Sci. Adv.* **3**(4), e1602531 (2017).
- ¹¹S. Datta and B. Das, *Appl. Phys. Lett.* **56**, 665–667 (1990).
- ¹²L. J. Cornelissen, J. Liu, B. J. van Wees, and R. A. Duine, *Phys. Rev. Lett.* **120**, 097702 (2018).
- ¹³A. A. Khudorozhkov, P. N. Skirdkov, K. A. Zvezdin, P. M. Vetoshko, and A. F. Popkov, *Phys. Rev. B* **96**, 214410 (2017).
- ¹⁴V. Balachandran, G. Benenti, E. Pereira, G. Casati, and D. Poletti, *Phys. Rev. Lett.* **120**, 200603 (2018).
- ¹⁵K. Olejník, V. Schuler, X. Marti, V. Novák, Z. Kaspar, P. Wadley, R. P. Campion, K. W. Edmonds, B. L. Gallagher, J. Garces, M. Baumgartner, P. Gambardella, and T. Jungwirth, *Nat. Commun.* **8**, 15434 (2017).
- ¹⁶X. Marti, I. Fina, C. Frontera, J. Liu, P. Wadley, Q. He, R. J. Paull, J. D. Clarkson, J. Kudrnovský, I. Turek, J. Kuneš, D. Yi, J.-H. Chu, C. T. Nelson, L. You, E. Arenholz, S. Salahuddin, J. Fontcuberta, T. Jungwirth, and R. Ramesh, *Nat. Mater.* **13**, 367 (2014).
- ¹⁷T. Jungwirth, X. Marti, P. Wadley, and J. Wunderlich, *Nat. Nanotechnol.* **11**, 231 (2016).
- ¹⁸O. Gomonay, T. Jungwirth, and J. Sinova, *Phys. Status Solidi RRL* **11**, 1700022 (2017).
- ¹⁹V. Baltz, A. Manchon, M. Tsoi, T. Moriyama, T. Ono, and Y. Tserkovnyak, *Rev. Mod. Phys.* **90**, 015005–1 (2018).
- ²⁰R. A. Duine, K. J. Lee, S. S. P. Parkin, and M. D. Stiles, *Nat. Phys.* **14**, 217 (2018).
- ²¹M. B. Jungfleisch, W. Zhang, and A. Hoffmann, *Phys. Lett. A* **382**, 865 (2018).
- ²²P. Němec, M. Fiebig, T. Kampfrath, and A. V. Kimel, *Nat. Phys.* **14**, 229 (2018).
- ²³L. Šmejkal, Y. Mokrousov, Y. Binghai, and A. H. MacDonald, *Nat. Phys.* **14**, 242 (2018).
- ²⁴J. Železný, P. Wadley, K. Olejník, A. Hoffmann, and H. Ohno, *Nat. Phys.* **14**, 220–228 (2018).
- ²⁵T. Jungwirth, J. Sinova, A. Manchon, X. Marti, J. Wunderlich, and C. Felser, *Nat. Phys.* **14**, 200 (2018).
- ²⁶F. L. A. Machado, P. R. T. Ribeiro, J. Holanda, R. L. Rodríguez-Suárez, A. Azevedo, and S. M. Rezende, *Phys. Rev. B* **95**, 104418 (2017).
- ²⁷R. Iguchi, A. Yagmur, Y.-C. Lau, S. Daimon, E. Saitoh, M. Hayashi, and K. Uchida, *Phys. Rev. B* **98**, 014402 (2018).
- ²⁸Y. Chen, D. Roy, E. Cogulu, H. Chang, M. Wu, and A. D. Kent, *Appl. Phys. Lett.* **113**, 202403 (2018).
- ²⁹P. R. T. Ribeiro, F. L. A. Machado, M. Gamino, A. Azevedo, and S. M. Rezende, *Phys. Rev. B* **99**, 094432 (2019).
- ³⁰S. M. Rezende, R. L. Rodríguez-Suárez, and A. Azevedo, *J. Phys. D: Appl. Phys.* **51**, 174004 (2018).
- ³¹T. Kikkawa, K. Uchida, Y. Shiomi, Z. Qiu, D. Hou, D. Tian, H. Nakayama, X.-F. Jin, and E. Saitoh, *Phys. Rev. Lett.* **110**, 067207 (2013).
- ³²C. T. Bui and F. Rivadulla, *Phys. Rev. B* **90**, 100403(R) (2014).